

Raspberry Pi 5 IMX415 Camera Driver Tutorial

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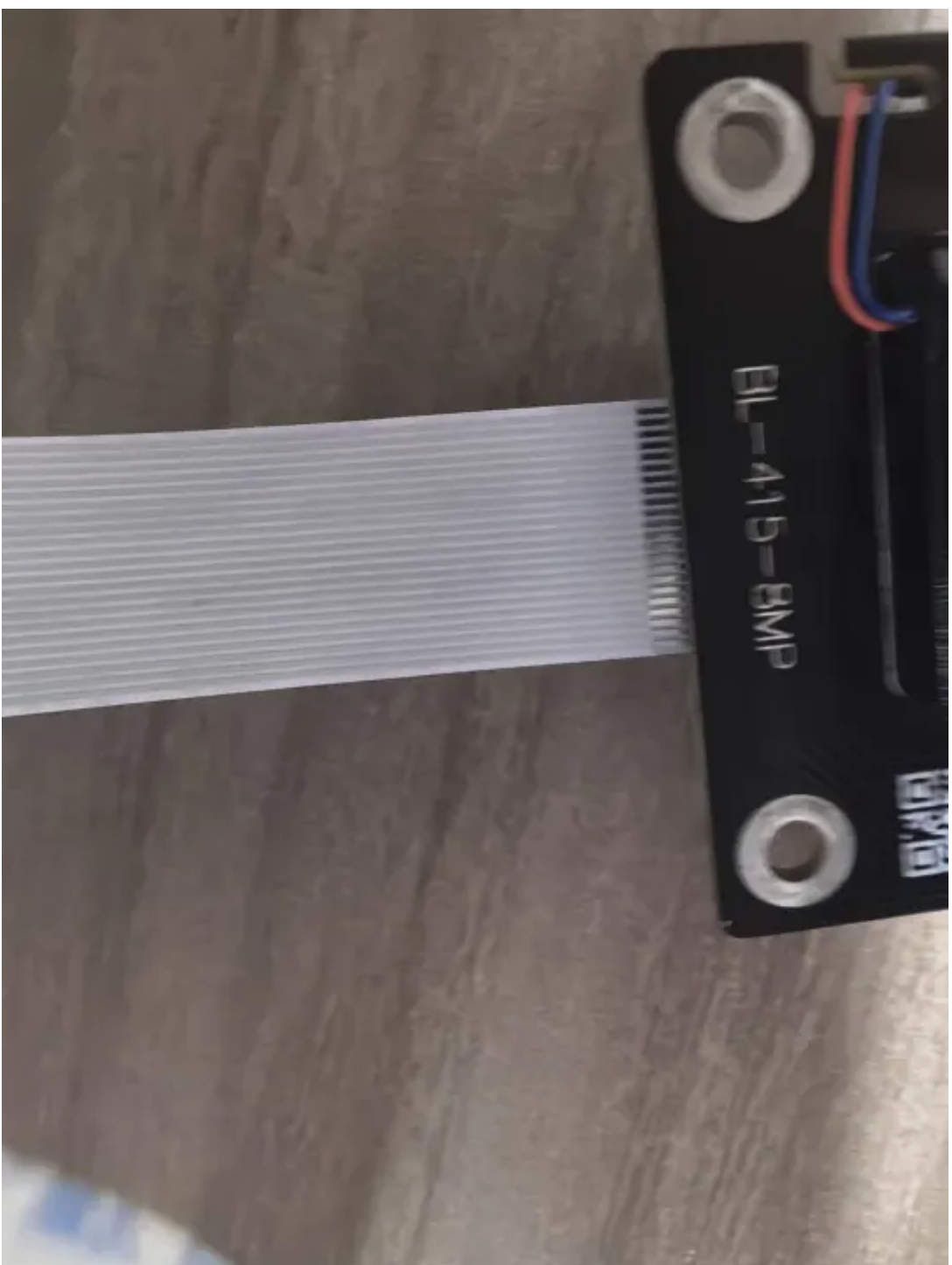
Friendly reminder: If you want to add a bracket, you need to insert all 4 screws at the same time first, and then tighten the screws.

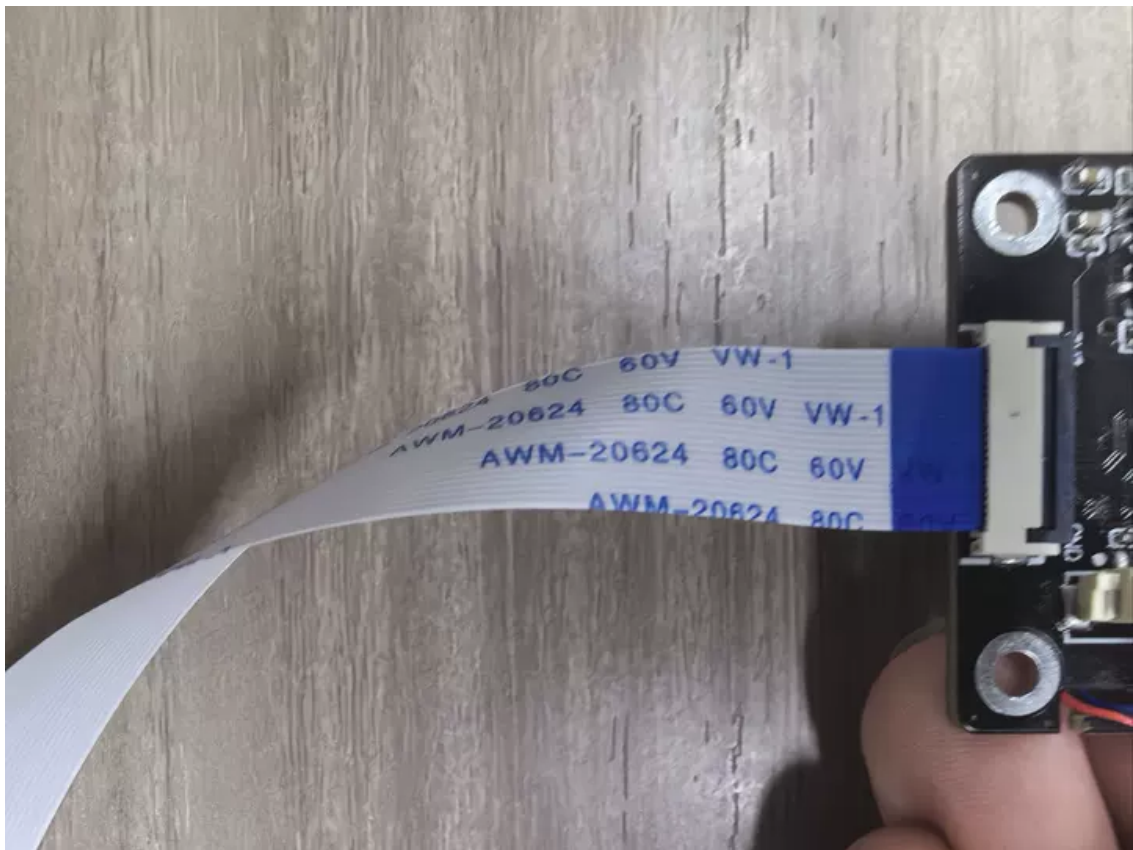
Notes:

- **Contact surface position:** The metal contacts at the end of the ribbon cable must face the spring contacts or contact array inside the FFC connector.
- **Blue stiffener:** The blue stiffener plate is usually located on the non-contact surface of the ribbon cable. In a standard installation, the metal contacts and the blue plate are usually located on opposite sides of the ribbon cable.
- **Silkscreen marking:** The plane where the silkscreen text on the ribbon cable (such as AWM-20624) is located is usually the back side. During installation, it should face away from the contact surface (the gold or silver metal part).

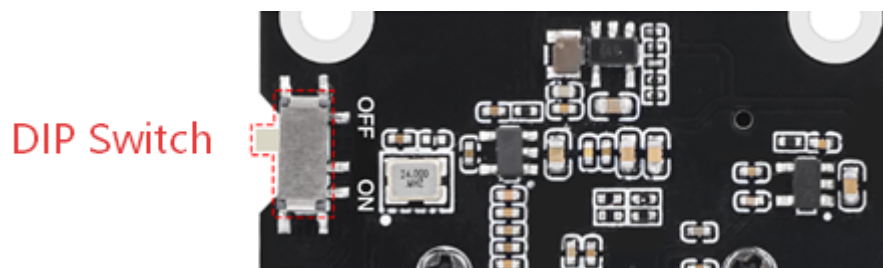
The metal must face the side against the PCB, otherwise the ribbon cable may burn out.

Then connect the other end to the metal contacts of the main board.





Note: On the IMX415 side, there is a DIP switch labeled IR-CUT, which is used to control the infrared function on and off. Set it to the ON position to turn on the infrared. It is normal for the screen to appear reddish. Set it to the OFF position to turn off the infrared function, and the screen will show normal colors.



1. Applicable Scope

This tutorial uses the kernel version of the latest official image as an example, namely the 6.18.34 kernel. The network drive also provides the imx415.ko and imx415.dtbo files compiled for the 6.6.31 and 6.12.25 kernels corresponding to our factory image. Check your corresponding kernel and download the corresponding kernel files for the operation.

This tutorial is applicable to:

- Raspberry Pi 5.
- Current kernel `6.18.34+rpt-rpi-2712`, source branch `rpi-6.18.y`.
- The IMX415 driver and overlay in the repository `pi5/src/`.
- The pre-compiled products for `6.18.34+rpt-rpi-2712` in `pi5/6.18.34/imx415/`.
- If your system is still `6.12.25+rpt-rpi-2712`, use `pi5/backup-rpi-6.12/` instead.

Repository file version relationship:

File	Kernel Version	Purpose
<code>pi5/6.18.34/imx415/src</code>	6.18 source	Compile and use on 6.18.34
<code>pi5/6.18.34/imx415/imx415.ko</code> and <code>pi5/6.18.34/imx415/imx415.dtbo</code>	6.18.34	Pre-compiled product for 6.18.34, can be installed directly
<code>pi5/6.12.25/imx415/</code>	6.12 source	Compile and use on 6.12.25
<code>pi5/6.12.25/imx415/imx415.ko</code> and <code>pi5/6.12.25/imx415/imx415.dtbo</code>	6.12.25	Pre-compiled product for 6.12.25, can be installed directly
<code>pi5/6.6.31/imx415/</code>	6.6.31	Compile and use on 6.6.31
<code>pi5/6.6.31/imx415/imx415.ko</code> and <code>pi5/6.6.31/imx415/imx415.dtbo</code>	6.6.31	Pre-compiled product for 6.6.31, can be installed directly

2. Hardware Description

- The IMX415 module connects to the MIPI CSI interface of the Pi 5.
- The default I2C address in the repository `pi5/src/imx415.dtsi` is `0x1a`.

3. Step 1: Confirm the Kernel Version

```
uname -r
```

Common version:

```
6.18.34+rpt-rpi-2712
```

If other versions are output, first confirm whether the current kernel matches the repository source. When the kernel version does not match, the driver may report `Invalid module format` or fail to load.

4. Method 1: Directly Use the Pre-compiled Product in the Repository

Only when `uname -r` is exactly `6.18.34+rpt-rpi-2712` can you directly use:

```
pi5/6.18.34/imx415/imx415.ko
pi5/6.18.34/imx415/imx415.dtbo
```

Copy the files to the Pi 5:

```
scp pi5/6.18.34/imx415/imx415.ko pi@<pi-ip>:~/
scp pi5/6.18.34/imx415/imx415.dtbo pi@<pi-ip>:~/
```

You can refer to the file transfer tutorial for the transfer. The purpose of this step is mainly to transfer the files to your main board for convenient operation. Transfer them to the corresponding directory, and the subsequent commands also need to be modified according to your actual path.

Install on Pi 5:

```
sudo mkdir -p /lib/modules/$(uname -r)/updates/dkms
sudo cp ~/imx415.ko /lib/modules/$(uname -r)/updates/dkms/
sudo depmod -a

sudo cp ~/imx415.dtbo /boot/firmware/overlays/
```

If the system uses the old path `/boot/overlays/`, copy to:

```
sudo cp ~/imx415.dtbo /boot/overlays/
```

Verify the module version:

```
modinfo imx415 | grep -E 'filename|vermagic'
```

Expected:

```
filename: /lib/modules/6.18.34+rpt-rpi-2712/updates/dkms/imx415.ko
vermagic: 6.18.34+rpt-rpi-2712 SMP preempt mod_unload modversions aarch64
```

5. Method 2: Compile on the Pi 5

It is recommended to compile yourself to avoid inconsistency between the pre-compiled product and the current kernel source.

5.1 Install Compilation Dependencies

```
sudo apt update
sudo apt install git make gcc bc bison flex libssl-dev device-tree-compiler \
raspberrypi-kernel-headers libcamera-dev libcamera-apps
```

Check whether the kernel header files are available:

```
ls /lib/modules/$(uname -r)/build
```

If `/lib/modules/$(uname -r)/build` does not exist, first install `raspberrypi-kernel-headers`, or download the complete kernel source.

5.2 Prepare the Source Code

On the Pi 5:

```
mkdir -p ~/imx415
cd ~/imx415
```

Copy the files under the repository `pi5/src/` to this directory:

```
imx415.c
imx415.dtsi
imx415-overlay.dts
Makefile
sync-from-kernel.sh
```

5.3 Compile the Kernel Module

```
make modules
```

Equivalent command:

```
make -C /lib/modules/$(uname -r)/build M=$(pwd) modules
```

After success, it generates:

```
imx415.ko
```

5.4 Compile the Device Tree Overlay

```
make dtbo
```

If `/lib/modules/$(uname -r)/build` lacks the complete DTS directory, you need to download the complete kernel source:

```
git clone --depth 1 --branch rpi-6.18.y \
https://github.com/raspberrypi/linux ~/linux
```

If your own Raspberry Pi differs from the one we provide, you can also directly download the files from <https://github.com/raspberrypi/linux> to obtain them.

```
linux/drivers/media/i2c/imx415.c
linux/arch/arm/boot/dts/overlays/imx415.dtsi
linux/arch/arm/boot/dts/overlays/imx415-overlay.dts
```

The Makefile and sync-from-kernel.sh must be in the same directory as imx415.c, imx415.dtsi, and imx415-overlay.dts.

Or you can go to this webpage to download the files of the corresponding version, and then transfer them to the main board.

Then use the complete kernel source path to compile the dtbo:

```
make dtbo KERNEL_SRC=/home/pi/linux
```

After success, it generates:

```
imx415.dtbo
```

5.5 Install the Compiled Products

```
sudo cp imx415.ko /lib/modules/6.18.34+rpt-rpi-2712/  
sudo cp imx415.ko /lib/modules/6.18.34+rpt-rpi-2712/kernel/drivers/media/i2c/  
sudo cp imx415.dtbo /boot/firmware/overlays/  
cd /lib/modules/6.18.34+rpt-rpi-2712/kernel/drivers/media/i2c/  
sudo xz -z imx415.ko  
sudo depmod -a
```

Confirm the module again:

```
modinfo imx415 | grep -E 'filename|vermagic'
```

The `vermagic` must contain the current `uname -r`.

6. Modify `/boot/firmware/config.txt`

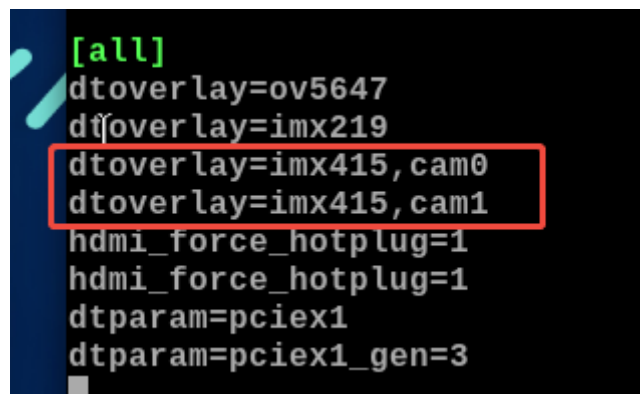
```
sudo nano /boot/firmware/config.txt
```

Single camera:

```
[all]  
dtoverlay=imx415,cam0
```

Dual camera:

```
[all]  
dtoverlay=imx415,cam0  
dtoverlay=imx415,cam1
```



```
[all]  
dtoverlay=ov5647  
dttoverlay=imx219  
dtoverlay=imx415,cam0  
dtoverlay=imx415,cam1  
hdmi_force_hotplug=1  
hdmi_force_hotplug=1  
dtparam=pciex1  
dtparam=pciex1_gen=3
```

After saving, reboot:

```
sudo reboot
```

7. Verify the Camera

After the operation and reboot, view the media devices:

```
ls -l /dev/video*
```

Test the image:

```
rpicam-hello -t 0
```

For multiple cameras, you can specify:

```
rpicam-hello -t 0 --camera 1
```

7.1 Preview the Camera

```
rpicam-hello
```

Entering this command in the terminal will display the preview window for about 5 seconds.

```
rpicam-hello -t 0
```

Running this command in the terminal will keep displaying the preview window. You can exit using the window close button and the Ctrl+C key!

7.2 Take a Photo

```
rpicam-jpeg -o test.jpg
```

Display a 5-second preview, then capture the image and save it as the test.jpg file.

```
rpicam-jpeg -o test.jpg -t 2000 --width 640 --height 480
```

Display a 2-second preview, then capture the image and save it as the test.jpg file. The width of the image is 640 pixels and the height is 480 pixels.

7.2.1 rpicam-still

This command can be used to save files in different formats:

```
rpicam-still -e png -o test.png  
rpicam-still -e bmp -o test.bmp  
rpicam-still -e rgb -o test.data  
rpicam-still -e yuv420 -o test.data
```

- Raw image capture

```
rpicam-still -r -o test.jpg
```

- Time-lapse shooting

Continuously capture images at an interval of 2 seconds, with a total capture duration of 30 seconds, and save each image with a filename format similar to image0001.jpg:

```
rpicam-still -t 30000 --timelapse 2000 -o image%04d.jpg
```

7.3 Video Recording

rpicam-vid

A command used to record video on the Raspberry Pi using a camera module.

Example: Record 10 seconds of video and write it to the test.h264 file.

```
rpicam-vid -t 10000 -o test.h264
```

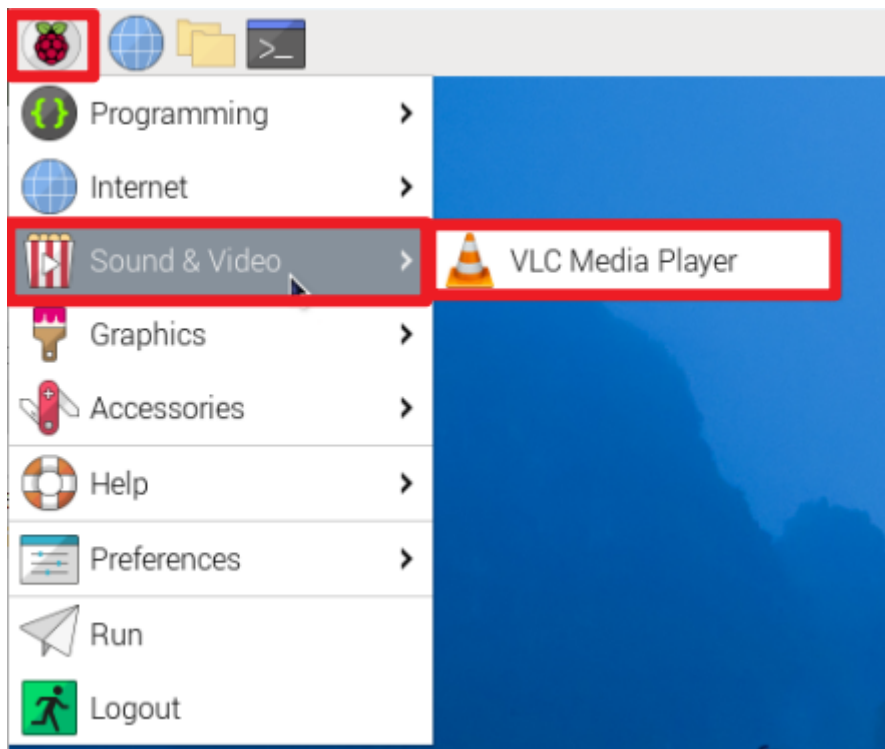
Play the video

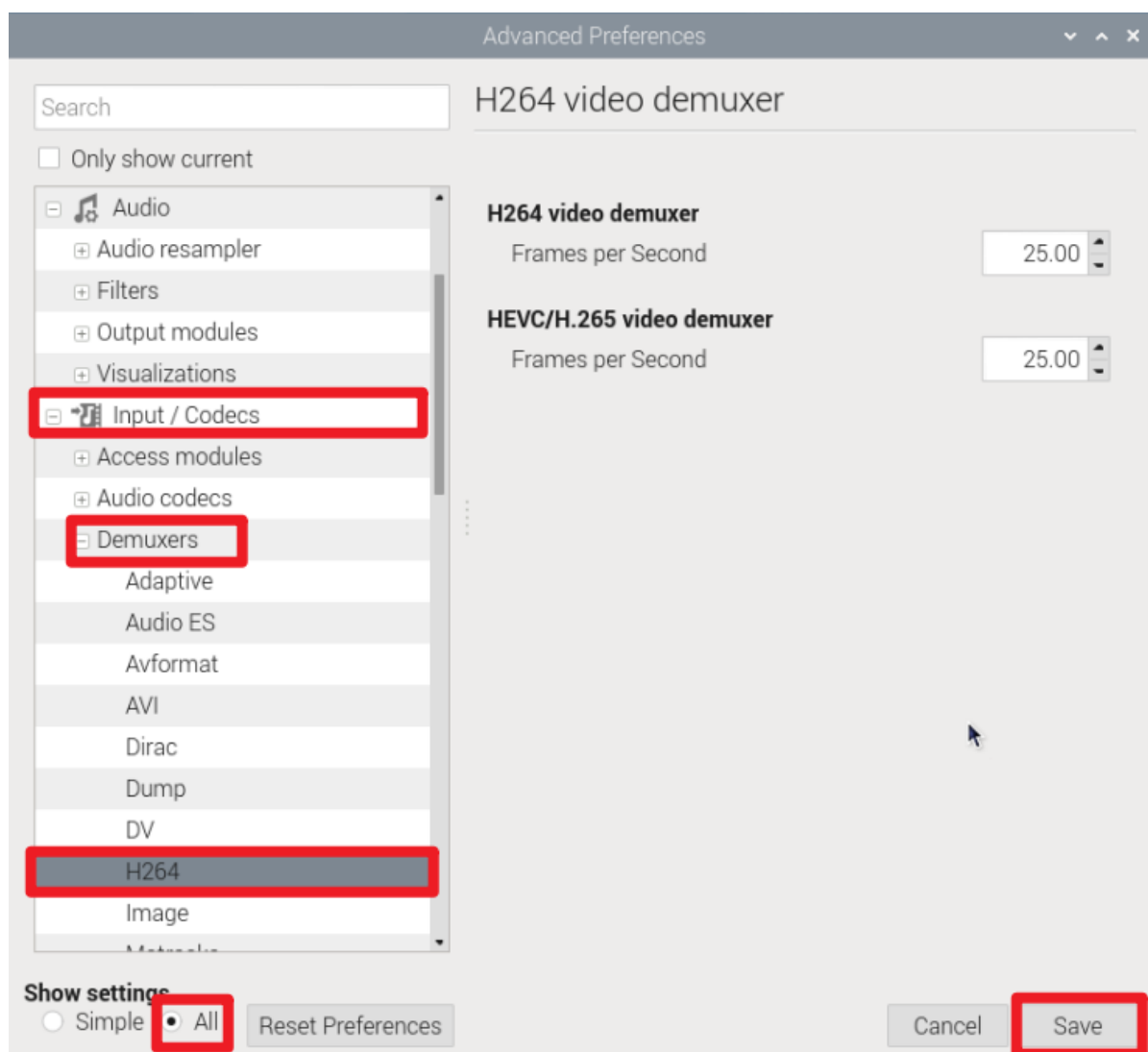
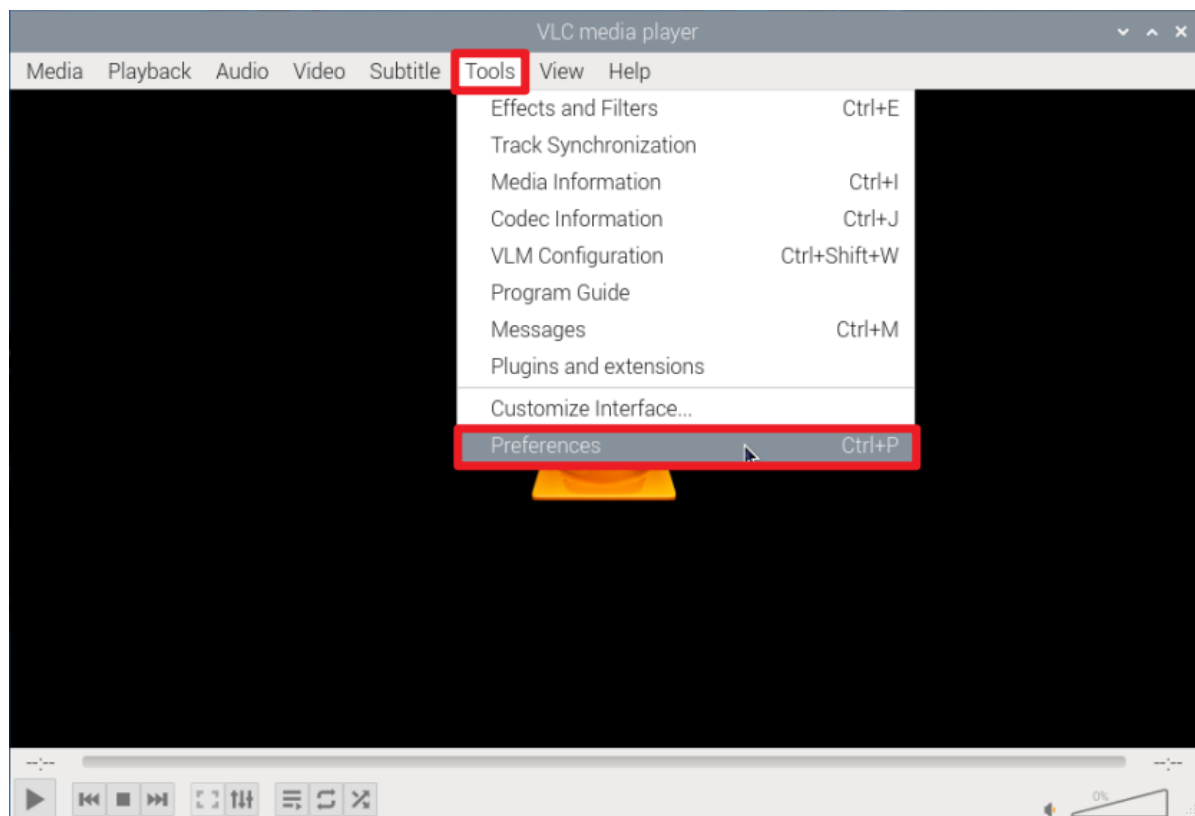
```
vlc test.h264
```

Note: If the test.h264 file cannot be played and an error is reported, please try the following method to solve it.

Error Solution

Modify the frame rate at which H264 plays per second.





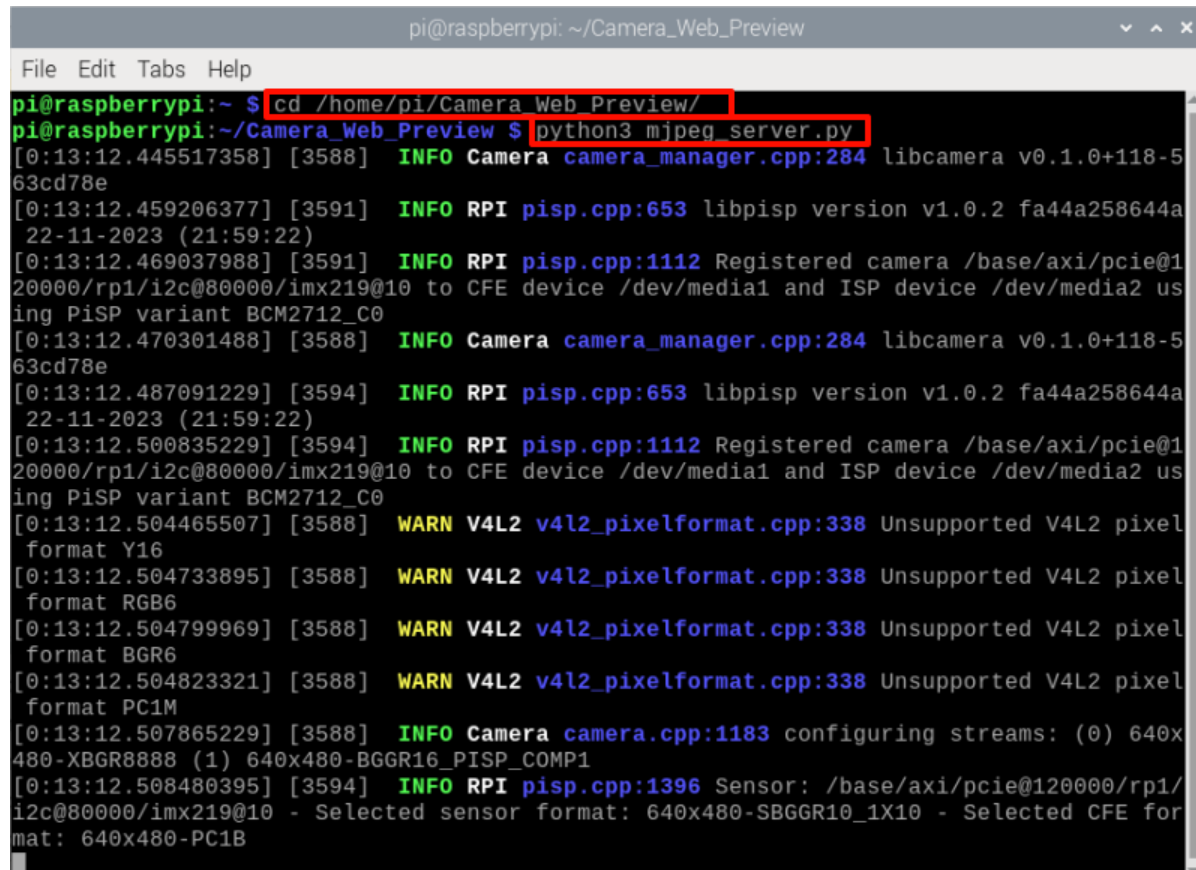
7.4 Web Page Camera Preview

Use a Python script file to implement a web page preview of the camera image.

7.4.1 Run the Script

Code path: /home/pi/Camera_Web_Preview/

```
cd /home/pi/Camera_Web_Preview/  
python3 mjpeg_server.py
```

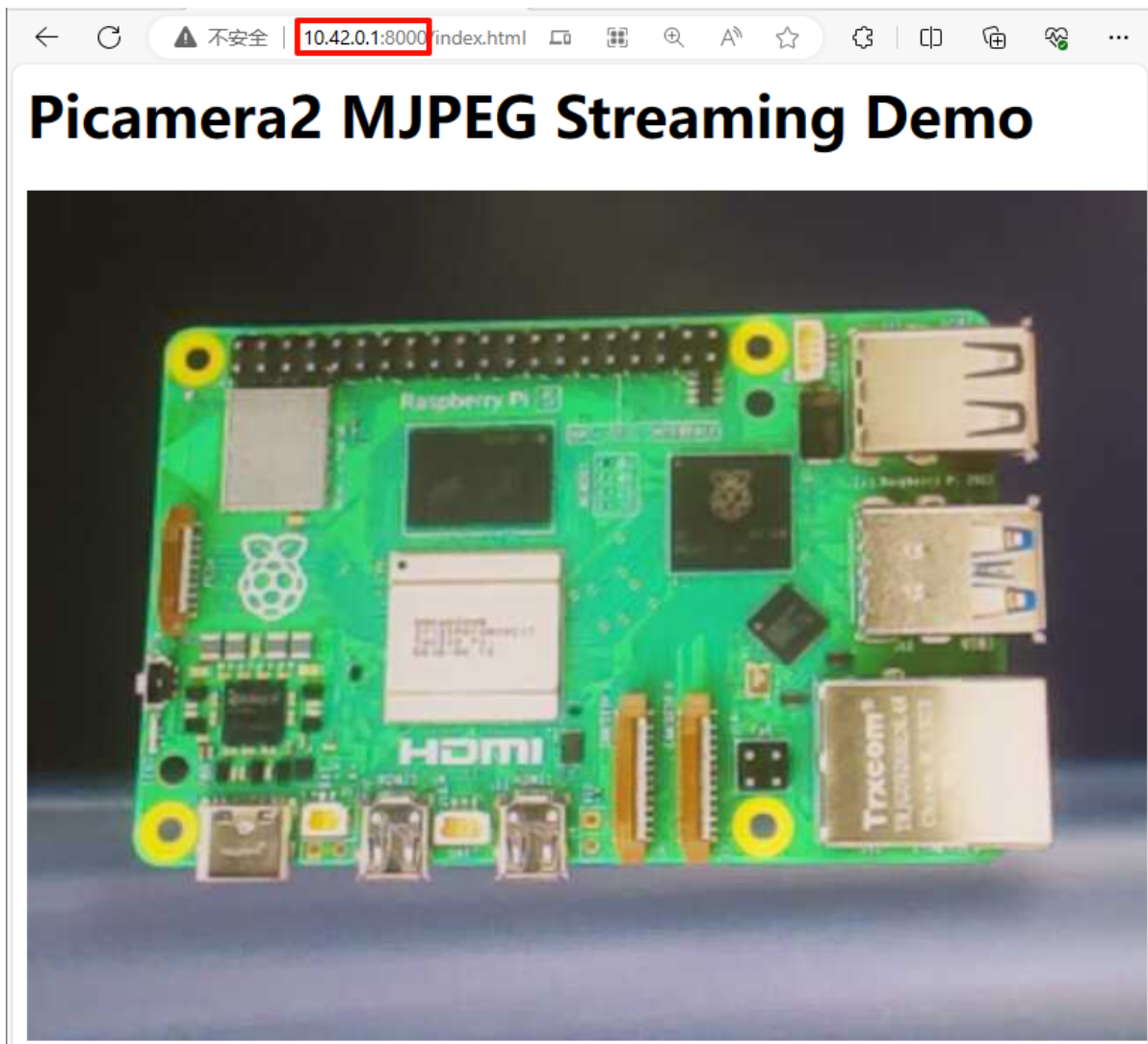


```
pi@raspberrypi:~/Camera_Web_Preview  
File Edit Tabs Help  
pi@raspberrypi:~$ cd /home/pi/Camera_Web_Preview/  
pi@raspberrypi:~/Camera_Web_Preview$ python3 mjpeg_server.py  
[0:13:12.445517358] [3588] INFO Camera camera_manager.cpp:284 libcamera v0.1.0+118-563cd78e  
[0:13:12.459206377] [3591] INFO RPI pisp.cpp:653 libpisp version v1.0.2 fa44a258644a22-11-2023 (21:59:22)  
[0:13:12.469037988] [3591] INFO RPI pisp.cpp:1112 Registered camera /base/axi/pcie@120000/rp1/i2c@80000/imx219@10 to CFE device /dev/media1 and ISP device /dev/media2 using PiSP variant BCM2712_C0  
[0:13:12.470301488] [3588] INFO Camera camera_manager.cpp:284 libcamera v0.1.0+118-563cd78e  
[0:13:12.487091229] [3594] INFO RPI pisp.cpp:653 libpisp version v1.0.2 fa44a258644a22-11-2023 (21:59:22)  
[0:13:12.500835229] [3594] INFO RPI pisp.cpp:1112 Registered camera /base/axi/pcie@120000/rp1/i2c@80000/imx219@10 to CFE device /dev/media1 and ISP device /dev/media2 using PiSP variant BCM2712_C0  
[0:13:12.504465507] [3588] WARN V4L2 v4l2_pixelformat.cpp:338 Unsupported V4L2 pixel format Y16  
[0:13:12.504733895] [3588] WARN V4L2 v4l2_pixelformat.cpp:338 Unsupported V4L2 pixel format RGB6  
[0:13:12.504799969] [3588] WARN V4L2 v4l2_pixelformat.cpp:338 Unsupported V4L2 pixel format BGR6  
[0:13:12.504823321] [3588] WARN V4L2 v4l2_pixelformat.cpp:338 Unsupported V4L2 pixel format PC1M  
[0:13:12.507865229] [3588] INFO Camera camera.cpp:1183 configuring streams: (0) 640x480-XBGR8888 (1) 640x480-BGGR16_PISP_COMP1  
[0:13:12.508480395] [3594] INFO RPI pisp.cpp:1396 Sensor: /base/axi/pcie@120000/rp1/i2c@80000/imx219@10 - Selected sensor format: 640x480-SBGGR10_1X10 - Selected CFE format: 640x480-PC1B
```

7.4.2 Web Page Access

Devices on the same local area network can enter `:8000` in a browser to view the real-time camera image!

```
Example:  
Raspberry Pi IP: 10.42.0.1  
web page access: 10.42.0.1:8000
```



If you want to set it to start automatically on boot, you can search for information online and set this script file to start automatically on boot!

8. 6.12 Kernel Backup Usage

If the current kernel of the Pi 5 is:

```
6.12.25+rpt-rpi-2712
```

Use `pi5/backup-rpi-6.12/`:

```
mkdir -p ~/imx415
cd ~/imx415
```

Copy the source code under `pi5/backup-rpi-6.12/`, then:

```
make modules
make dtbo
```

If the headers lack DTS:

```
git clone --depth 1 --branch rpi-6.12.y https://github.com/raspberrypi/linux
~/linux
make dtbo KERNEL_SRC=/home/pi/linux
```

The installation and configuration steps are the same as for 6.18.

9. Re-adaptation When Switching Other Raspberry Pi Kernels

Prepare a complete kernel source that matches the current kernel, then:

```
cd ~/imx415
make modules KERNEL_SRC=/home/pi/linux
make dtbo
```

Note: `pi5/src/imx415.c` has been adapted to the `devm_v4l2_sensor_clk_get` interface of 6.18. If you sync from the 6.12 source to the 6.18 kernel, you need to refer to the driver in the 6.18 kernel, and you cannot directly mix the old source.

10. Common Problems

10.1 After installation, `modinfo imx415` shows the old path or the old version

Execute:

```
modinfo imx415 | grep -E 'filename|vermagic'
```

If it shows `6.6.31` or `6.12.25`, it means the system loaded the old module. Delete the old file and install the matching version:

```
sudo rm -f /lib/modules/$(uname -r)/updates/dkms/imx415.ko
sudo cp imx415.ko /lib/modules/$(uname -r)/updates/dkms/
sudo depmod -a
```

10.2 An error of `Invalid module format` or `disagrees about version of symbol` is reported when loading

The kernel module does not match the current kernel version. Recompile using the source corresponding to the current `uname -r`.

10.3 The overlay does not take effect

Check:

```
ls -l /boot/firmware/overlays/imx415.dtbo
cat /boot/firmware/config.txt | grep dtoverlay
dmesg | grep -i overlay
```

You must reboot after modifying `config.txt`.

11. Notes

- Do not directly install `pi5/imx415.ko` or `pi5/imx415.dtbo` on 6.18.34. Download or compile the kernel files of the corresponding version and install them onto the corresponding version.
- After a kernel upgrade, `imx415.ko` and `imx415.dtbo` must be recompiled.
- After installing to `/lib/modules/$(uname -r)/updates/dkms/`, you must execute `sudo depmod -a`.
- The dtbo path should be selected as `/boot/firmware/overlays/` or `/boot/overlays/` according to the actual boot partition of the system.